

Sales Data Warehouse

Data Catalog & Project References

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Project Overview:

A SQL Server-based Sales Data Warehouse that transforms raw sales data into clean, analytics-ready datasets using the **Bronze, Silver, and Gold Medallion Architecture**. The project focuses on ETL, data quality, data modeling, and building a Sales Data Mart for analytical reporting.

Business Reequipments:

The business requires a SQL Server-based Sales Data Warehouse to consolidate, clean, and transform sales data into a reliable structure for analysis and reporting.

The solution must meet the following requirements:

1- **Sales Tracking:** Track each order at the individual product level, allowing multiple products within the same order to be analyzed separately.
2- **Order Lifecycle Analysis:** Track order dates, shipping dates, and due dates to analyze order fulfillment and delivery timelines.
3- **Sales Performance Analysis:** Analyze sales amounts, quantities sold, and product prices across different products and categories.
4- **Profitability Analysis:** Calculate and analyze product costs, profits, and profit margins to evaluate business performance.
5- **Tax Analysis:** Calculate the value-added tax (VAT) amount and percentage for sales transactions.
6- **Returns Analysis:** Identify returned transactions and evaluate their impact on sales and profitability.
7- **Product Analysis:** Analyze sales by product, category, subcategory, product line, and product cost, while considering product validity periods and historical changes.
8- **Customer Analysis:** Analyze sales by customer, country, gender, and marital status to understand customer purchasing behavior.
9- **Data Quality:** Identify and handle missing values, invalid dates, inconsistent sales amounts, incorrect prices, and other data issues that may affect analytical results.
10- **Data Integration:** Integrate sales, product, and customer data through appropriate fact and dimension relationships, ensuring accurate and consistent reporting.
11- **Analytics-Ready Data:** Build a Sales Data Mart that supports business reporting, KPI calculations, and data-driven decision-making.
12- **Data Validation:** Validate transformed data, record counts, calculations, and relationships across the data warehouse layers to ensure data accuracy and consistency.

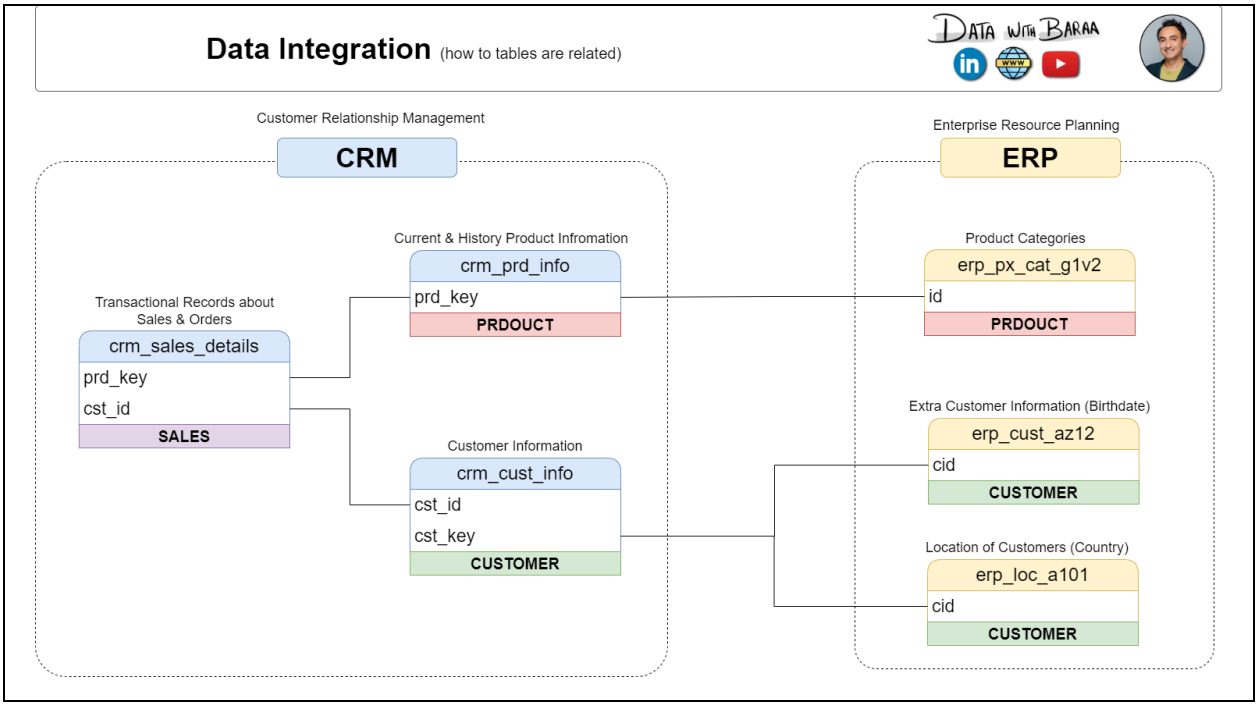
Data Sources:

Data Type: CSV Files

Description:
The project uses two source systems, CRM and ERP, each containing CSV files that provide customer, product, and sales data required to build the Sales Data Warehouse.

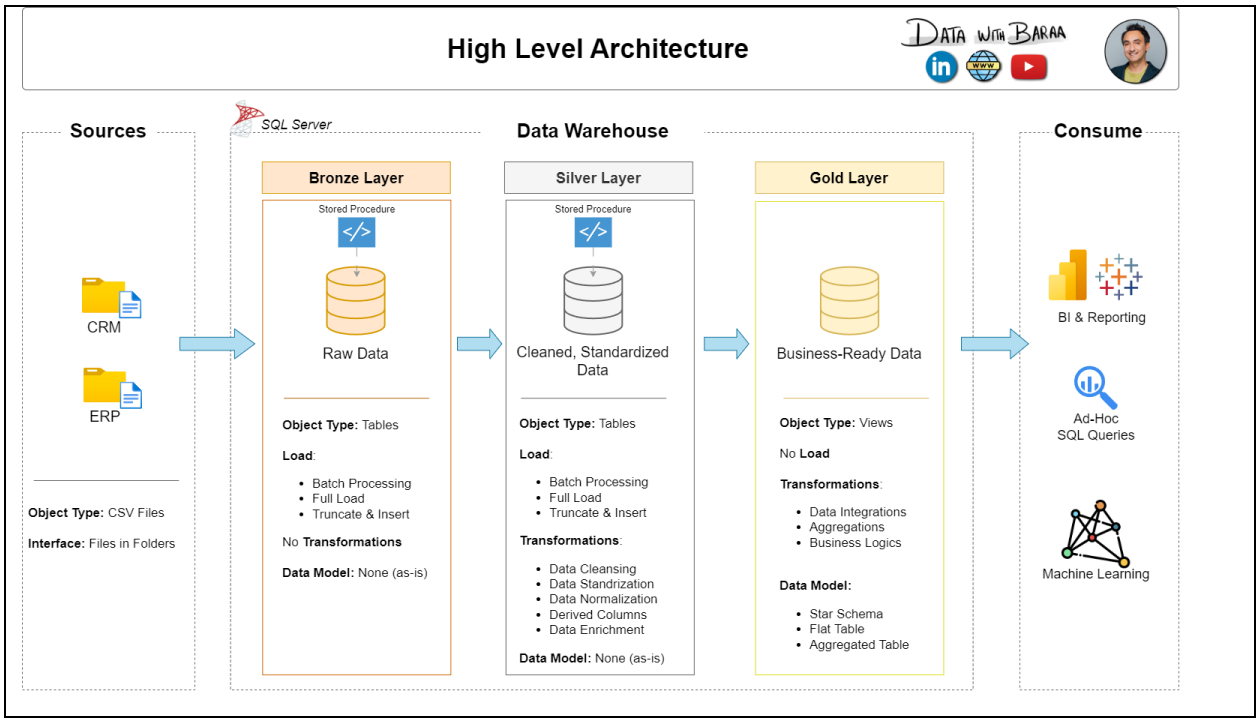
- **CRM (Customer Relationship Management):** Contains customer information and sales transaction details.
- **ERP (Enterprise Resource Planning):** Contains product information, product categories, and related business data.

Data Integrations (How tables are related):

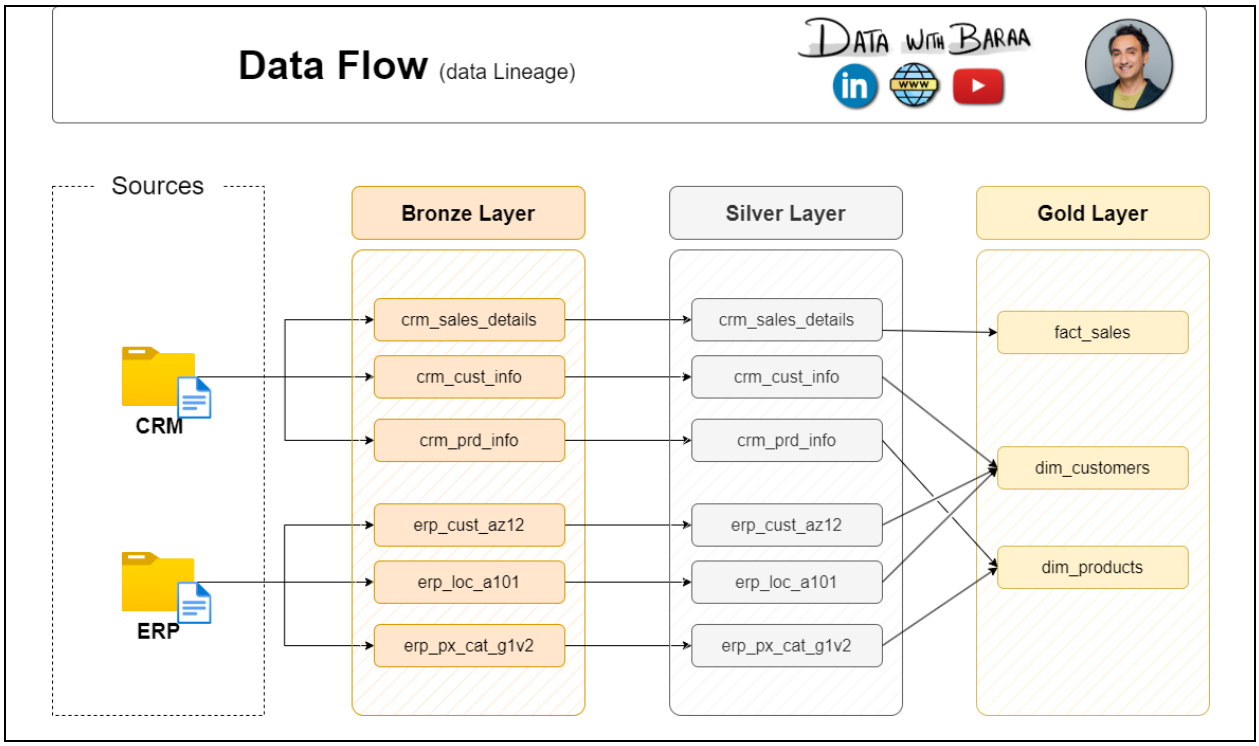


Data Architecture:

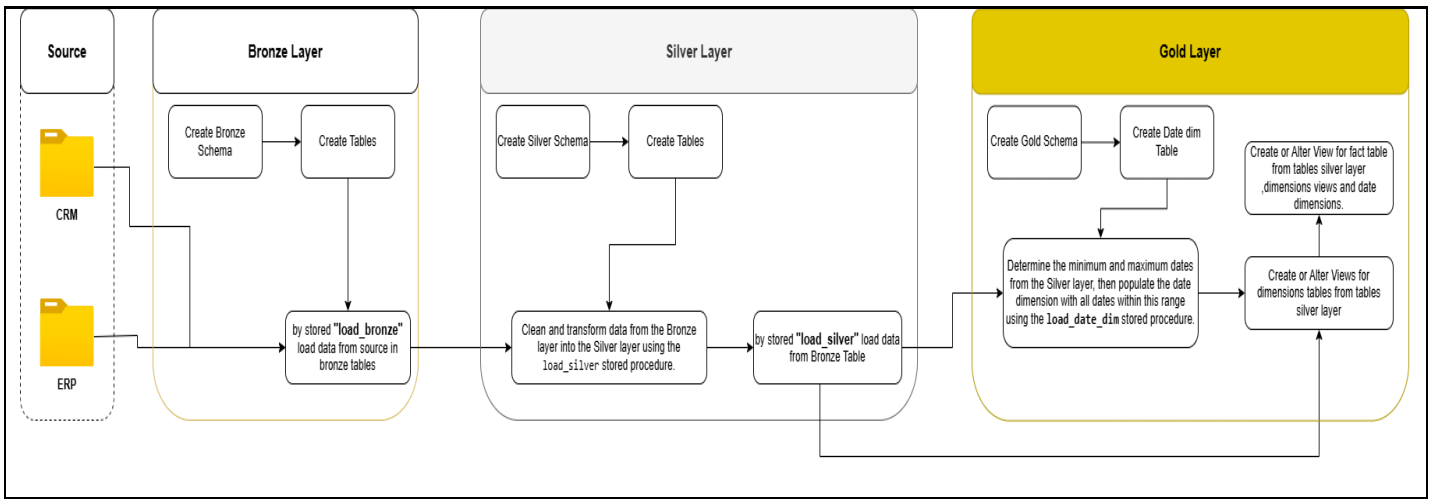
- Architecture Diagram:



- Data Flow:



ETL Pipeline:



Data Modeling:

Grain:
One row per product within each order, representing the sales transaction at the individual product level.

Fact Table:

- fact_sales: Stores sales transaction measures, including sales amount, quantity, price, profit, VAT amount, VAT percentage, and return indicator.

Dimension Tables:

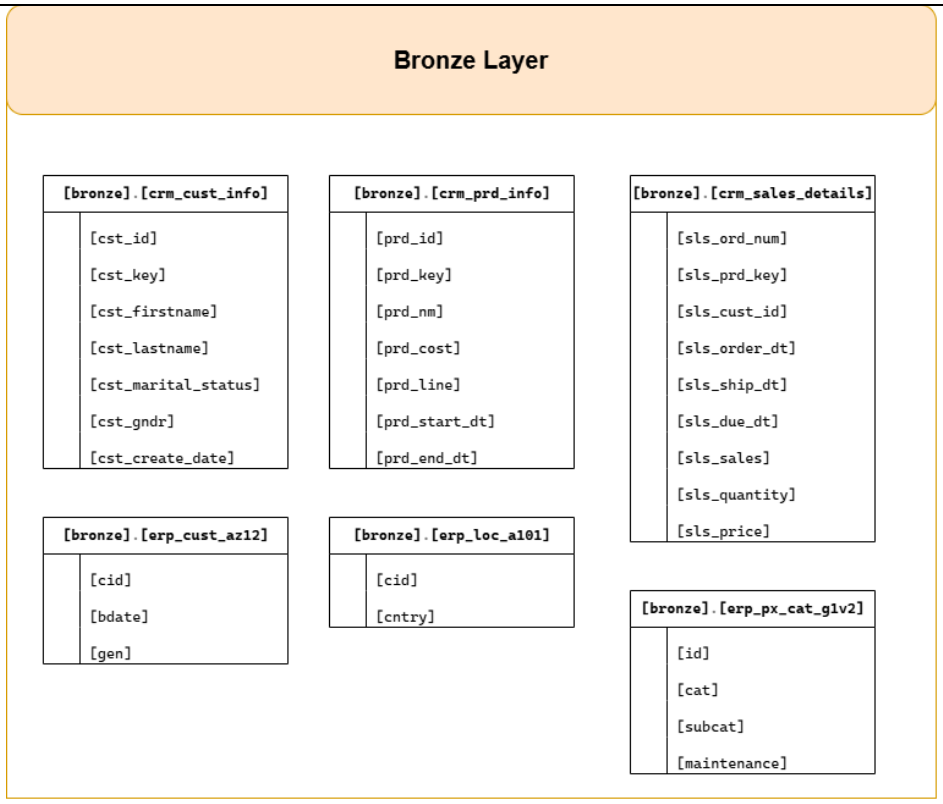
- dim_products: Contains product details, categories, subcategories, costs, product lines, and historical validity periods.
- dim_customers: Contains customer details, including name, country, gender, marital status, and birthdate.
- dim_date: Provides date attributes for analyzing order, shipping, and due dates.

Data Layers:

- Bronze Layer:
  - Data Dictionary:

- Stores raw data from CRM and ERP CSV files without applying any transformations or modifications.
- Since the source data contains inconsistencies and data quality issues, integrating tables directly at this stage is challenging and may lead to inaccurate relationships.

Tables:



Silver Layer:

Data Dictionary:

- Source Column: Original column name in the source.
- Target Column: Destination column name in Silver.
- Data Type: Actual data type in the target table.

- **Based on the business requirements, profit is calculated using the product cost effective at the order date. If no historical product pricing period matches the order date, the latest available product cost is used to calculate profit.**

Note: Columns not mentioned below are transferred from the Bronze layer without any changes.

Table: silver.crm_cust_info

Note: Keeps only the latest record per customer based on cst_create_date.

Column	Transformation
cst_firstname, cst_lastname	Removes leading and trailing spaces using TRIM().
cst_marital_status	Maps S to Single, M to Married, and other values to n/a.
cst_gndr	Maps F to Female, M to Male, and other values to n/a.

Table: silver.crm_prd_info

Column	Transformation
prd_key	Extracts the product identifier and replaces _ with -.
cat_key	Extracts the category key and replaces - with _.
prd_cost	Replaces NULL with 0.
prd_line	Maps product line codes to descriptive names; unknown values become n/a.
prd_end_dt	Derives the end date from the next product version's start date minus one day. The latest version retains NULL.

Table: silver.crm_sales_details

Column	Transformation
prd_id	Retrieves the matching historical product ID or uses the latest product ID as a fallback.
sls_order_dt	Converts valid dates to DATE; invalid values become 1900-01-01.
sls_ship_dt	Converts valid dates to DATE; invalid values are replaced with the order date minus one day.
sls_due_dt	Converts valid dates to DATE; invalid values are replaced with the shipping date minus one day.
sls_sales	Recalculates sales as sls_quantity * ABS(sls_price) when missing or inconsistent.

sls_price	Corrects missing or non-positive prices using sls_sales / sls_quantity under the applicable conditions.
is_return	Sets 1 if the original sales amount or price is negative; otherwise, 0.
tax_amount	Calculates VAT using the 14% rate and rounds to two decimal places.
tax_percentage	Calculates the VAT percentage relative to the price before VAT.
profit	Calculates unit profit as the corrected price before VAT minus the historical or fallback product cost.
prd_source_type	Identifies the cost source as Historical or Latest_Fallback.
prd_source_id	Stores the fallback product ID when historical cost is unavailable; otherwise, NULL.

Table: silver.erp_cust_az12

Column	Transformation
cid	Removes the NAS prefix when present.
bdate	Replaces future birth dates with NULL.
gen	Standardizes gender values to Female, Male, or n/a.

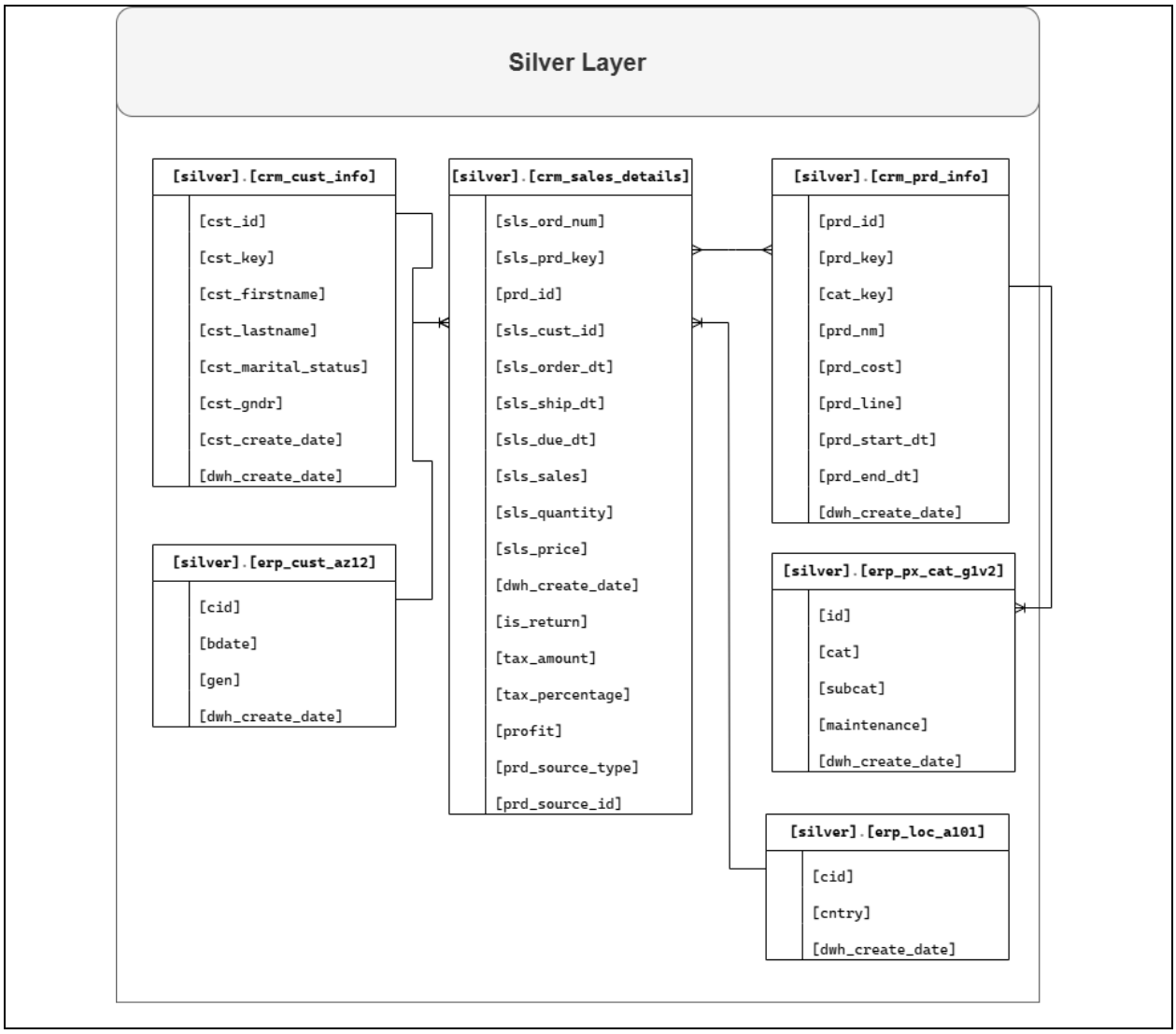
Table: silver.erp_loc_a101

Column	Transformation
cid	Removes hyphens (-) from the customer ID.
cntry	Maps DE to Germany, US/USA to United States, and blank or NULL values to n/a; trims other values.

Table: silver.erp_px_cat_g1v2

No column-level transformations are applied. All columns are transferred from Bronze as they are.

- Tables:



- Gold Layer:
  - Data Dictionary:

1. gold.dim_customers

Column	Description
customer_key	Surrogate key generated using ROW_NUMBER() ordered by cst_id.

customer_id	Original customer ID.
customer_number	Customer business key.
first_name	Customer first name.
last_name	Customer last name.
birthdate	Customer birthdate from ERP data.
country	Customer country from ERP data.
marital_status	Customer marital status: Single, Married, or n/a.
gender	Uses CRM gender if not n/a; otherwise uses ERP gender. Values: Female, Male, n/a.
create_date_key	Date key corresponding to the customer's creation date.

2. gold.dim_products

Column	Description
product_key	Surrogate key generated using ROW_NUMBER() ordered by prd_key.
product_id	Original product ID.
product_number	Product business key.
product_name	Product name.
category_id	Product category identifier.
category	Product category name from ERP data.
sub_category	Product subcategory from ERP data.
product_cost	Product cost; NULL values are replaced with 0 in Silver.
product_line	Product line classification: M → Mountain, R → Road, S → Other Sales, T → Touring, otherwise n/a.
product_start_date_key	Date key corresponding to the product version's start date.
product_end_date_key	Date key corresponding to the product version's end date; NULL end dates use 1900-01-01.
maintenance	Maintenance classification from ERP category data.

3. gold.fact_sales

Grain: One row per product within an order.

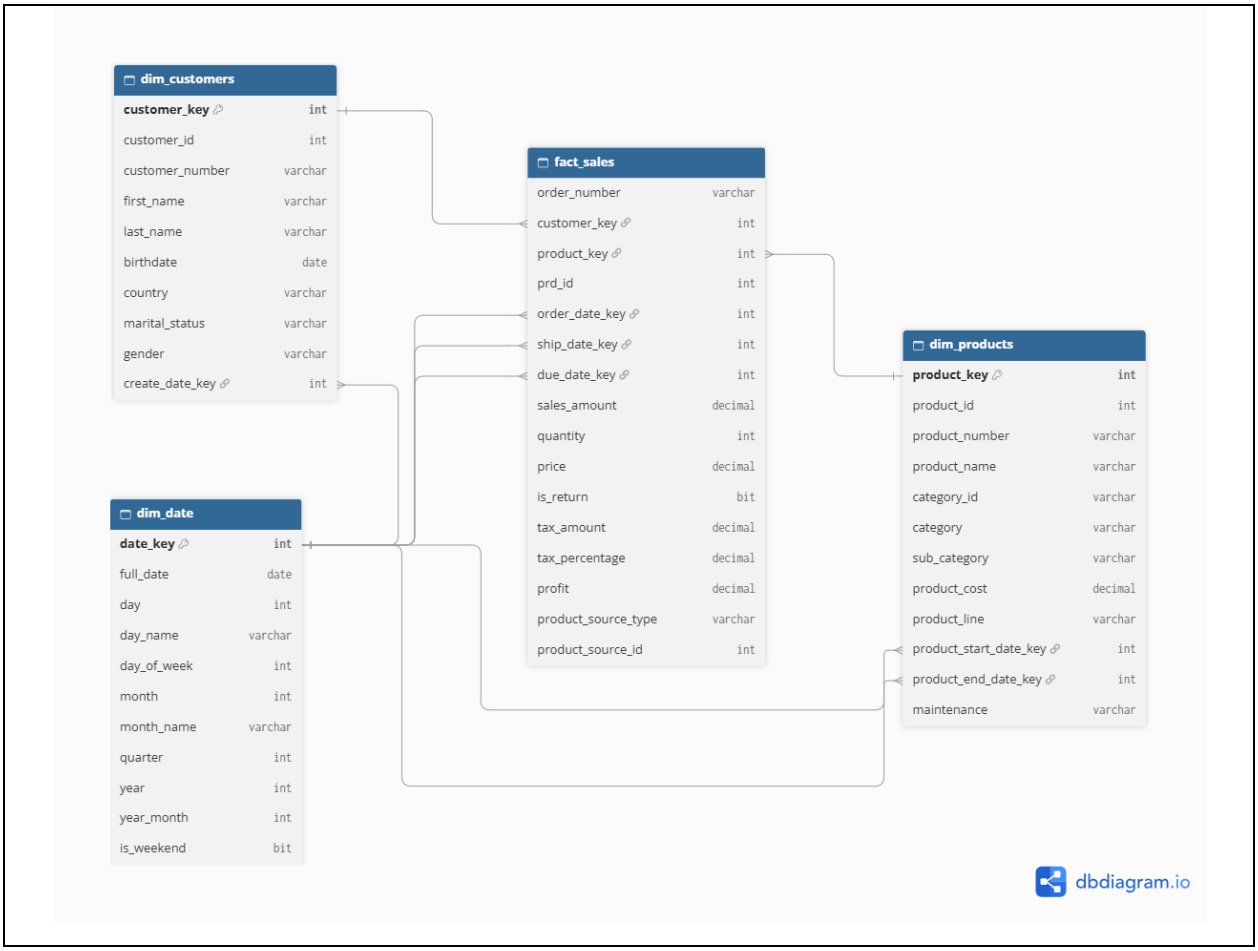
Column	Description
order_number	Sales order identifier.
customer_key	Surrogate key linking the transaction to gold.dim_customers.
product_key	Surrogate key linking the transaction to gold.dim_products.
prd_id	Product ID from the Silver Layer.
order_date_key	Date key corresponding to the order date.
ship_date_key	Date key corresponding to the shipping date.
due_date_key	Date key corresponding to the due date.
sales_amount	Sales amount for the transaction line; recalculated in Silver when NULL or inconsistent with quantity and price.
quantity	Number of units sold.
price	Corrected unit price.
is_return	Return indicator: 1 if the original sales amount or price was negative; otherwise 0.
tax_amount	Calculated tax amount based on the unit price, rounded to 2 decimal places.
tax_percentage	Calculated tax percentage based on the net unit price, rounded to 2 decimal places.
profit	Unit profit calculated using the product cost effective at the order date. If no historical pricing period matches, the latest available product cost is used.
product_source_type	Indicates the cost source: Historical or Latest_Fallback.
product_source_id	Fallback product ID when historical product cost is unavailable; otherwise NULL.

4.gold.dim_date

Description: The Date Dimension was generated based on the minimum and maximum dates identified across the source data, covering the entire period from the year of the earliest date to the year of the latest date.

Column	Description
date_key	Unique date identifier (Primary Key).
full_date	Complete calendar date.
day	Day of the month.
day_name	Name of the day.
day_of_week	Numeric day of the week.
month	Month number (1–12).
month_name	Name of the month.
quarter	Calendar quarter (1–4).
year	Calendar year.
year_month	Year and month in YYYYMM format.
is_weekend	Indicates whether the date is a weekend.

- Data Modeling:



Technologies Used:

- **SQL Server:** Database management and data warehouse implementation.
- **T-SQL:** Data cleaning, transformation, business rules, and stored procedures.
- **SQL Server Management Studio (SSMS):** Database development, query execution, and data validation.

Links & References:

- **Project Implementation Tutorials:** https://youtube.com/playlist?list=PLNcg_FV9n7qaUWeyUkPfiVtMbKlrfMqA8&si=ZR5IiNx6o17QA0gE
- **LinkedIn:** www.linkedin.com/in/roqiua-mohammad
- **GitHub Repository:** https://github.com/roqiua/sql_data_warehouse_case_study